Lunar Surface Fission Power (LSFP) System

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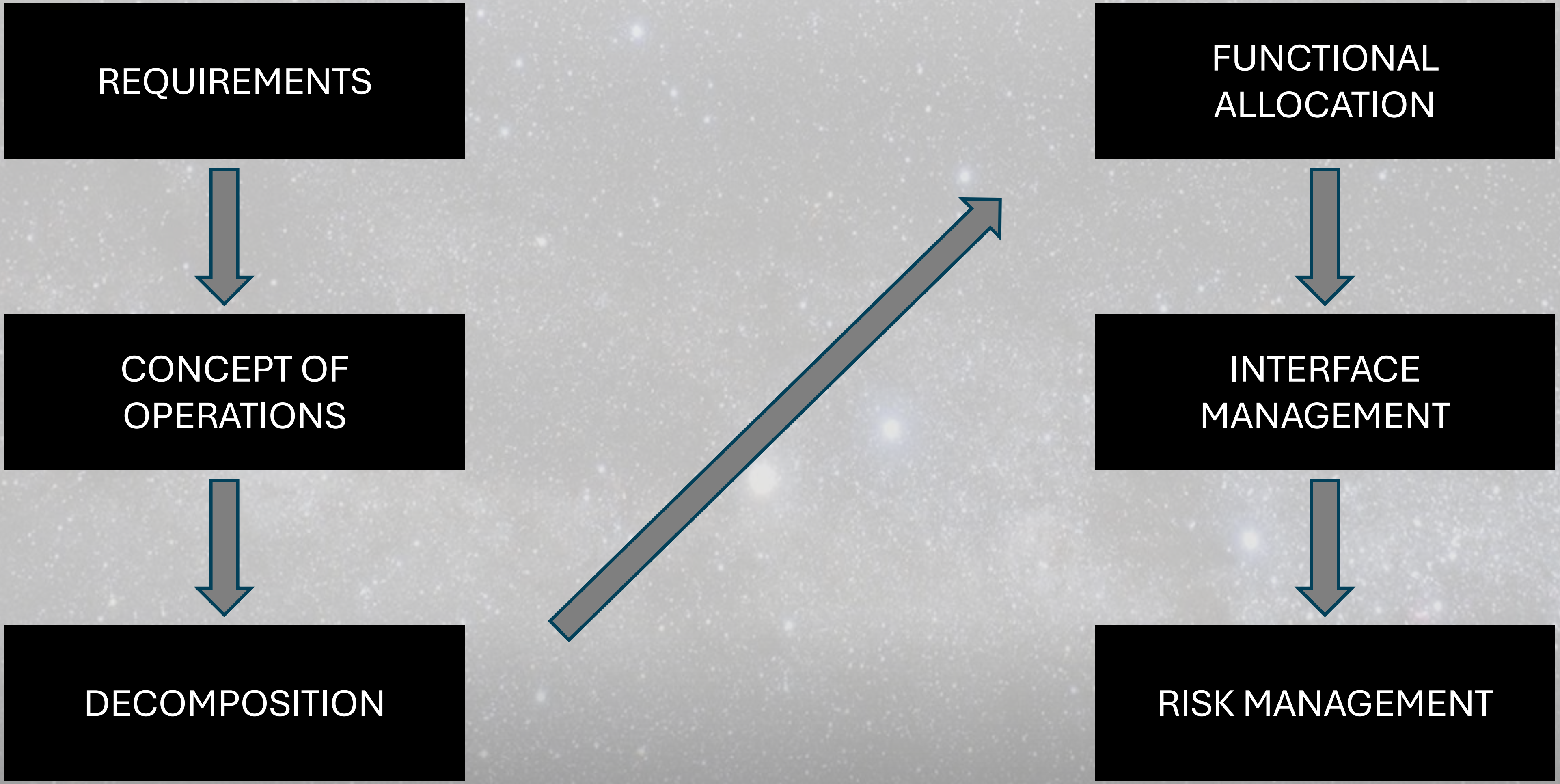
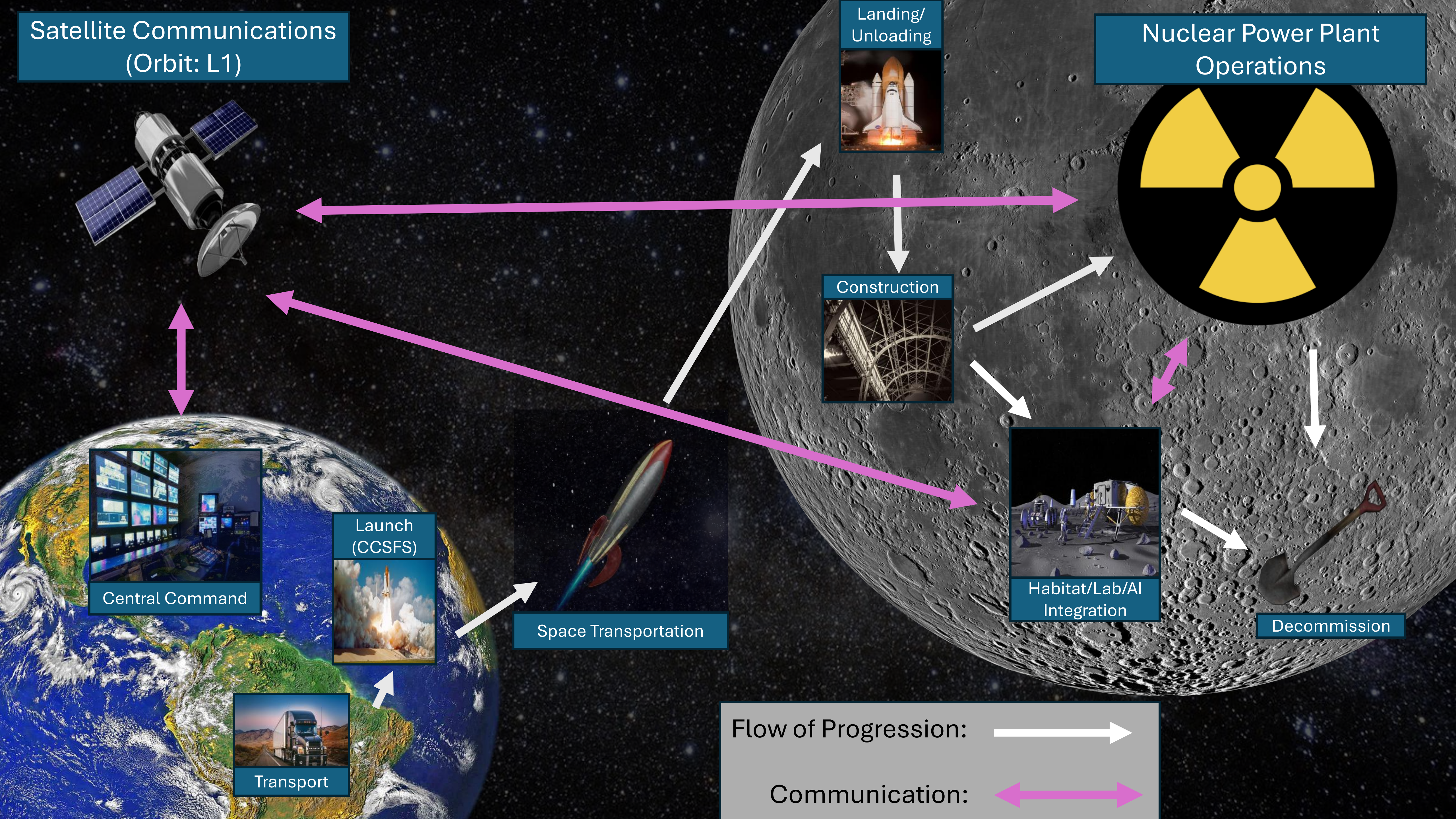
College of Engineering

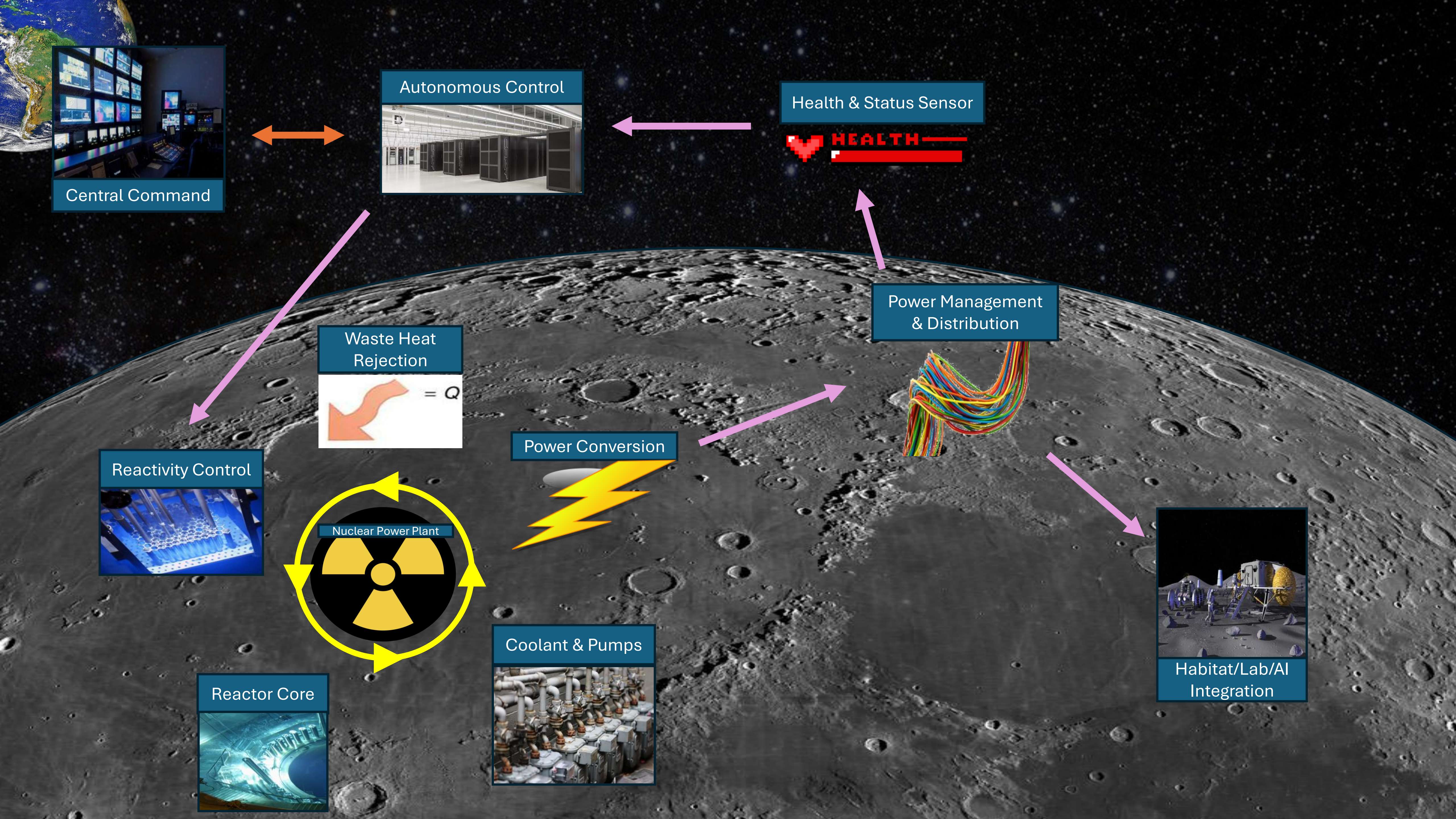
Systems Engineering Approach



Concept of Operations



Satellite Communications (Orbit: L1)
Landing/ Unloading
Nuclear Power Plant Operations
Construction
Central Command
Launch (CCSFS)
Space Transportation
Habitat/Lab/AI Integration
Decommission
Transport
Flow of Progression:
Communication:



Central Command
Autonomous Control
Health & Status Sensor
HEALTH
Power Management & Distribution
Waste Heat Rejection
= Q
Power Conversion
Reactivity Control
Nuclear Power Plant
Coolant & Pumps
Reactor Core
Habitat/Lab/AI Integration

Design Drivers

In the scope of this mission, **three** design drivers stood out:

1. **Mission Life Span & Reliability**

2. **Reactor System Properties (Size, Mass, Power)**

3. **Power Distribution (PMAD)**

Mission Life Span & Reliability

DRIVER	10-year autonomous operation in a harsh environment.
RISK	Latent faults, radiation upsets, wear-out under thermal cycling
MITIGATION	Fail-safe/reactivity shutdown, diverse sensors with voting + health monitoring, rad-hard/derated electronics, TVAC/dust life-testing

Reactor System Properties (Size, Mass, Power)

DRIVER

Meet ~40 kWe (scalable to ~100 kWe) within single-launch (SLS/Starship) envelope

Scalability achieved through continued advancements in AI-driven control systems and planned hardware upgrades to support higher power outputs.

RISK

Exceed mass/volume; insufficient specific power; oversized radiators/shielding; landing/deployment issues.

MITIGATION

High-specific-power core + efficient conversion (Brayton/Stirling), deployable mass-optimized radiators, directional/shadow shielding + standoff, packaging for CG/shock margins.

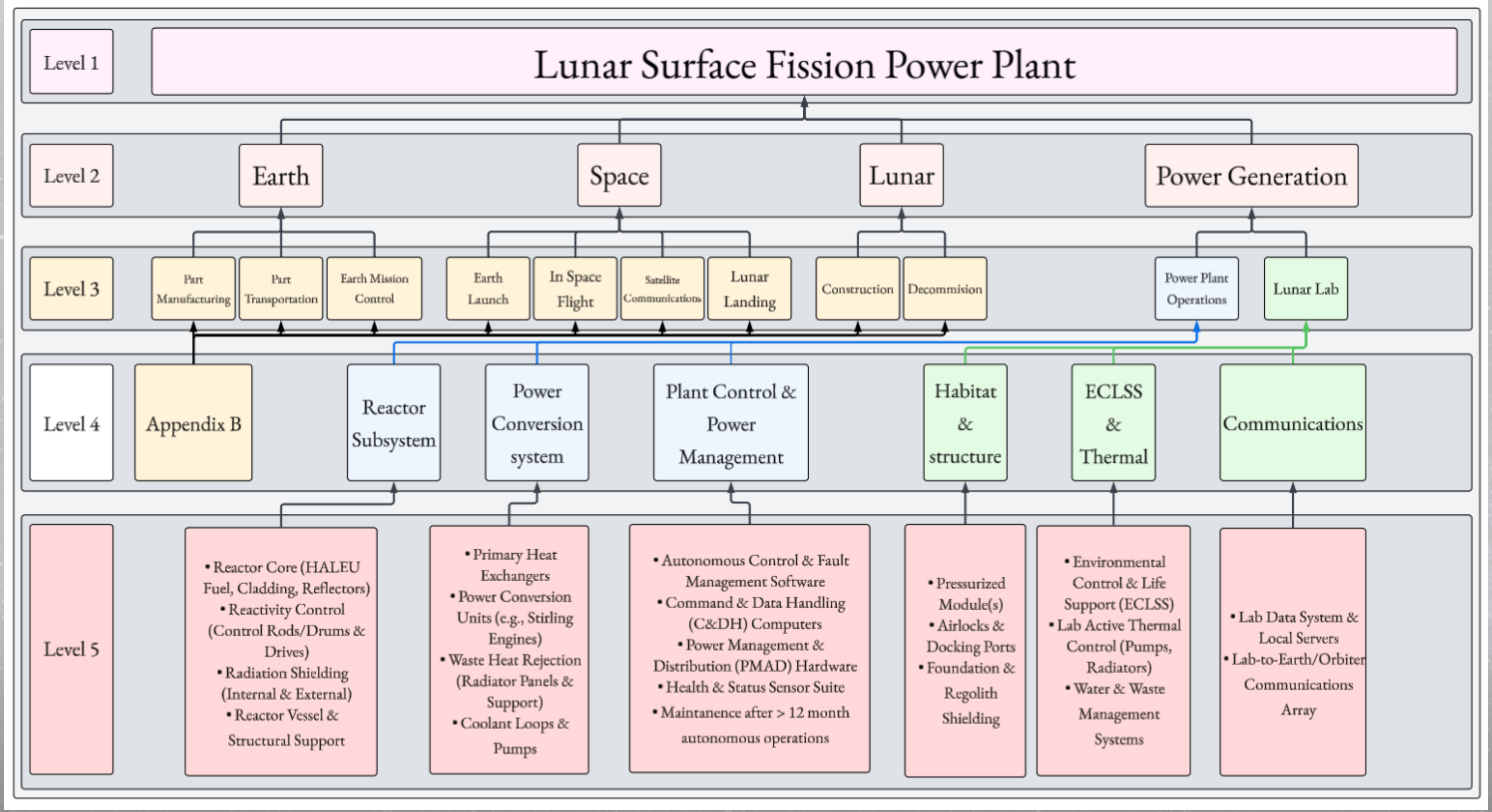
Power Distribution (PMAD)

DRIVER	Efficient, reliable delivery from standoff site to habitat/ISRU with stable power quality.
RISK	Heat dissipation losses and faults over long runs; regulation/harmonics; dust/thermal damage to cables/connectors
MITIGATION	High-voltage AC/DC backbone with point-of-load conversion, coordinated protection/sectionalizing + autonomous load shedding, hardened cabling/sealed terminations with protected routing.

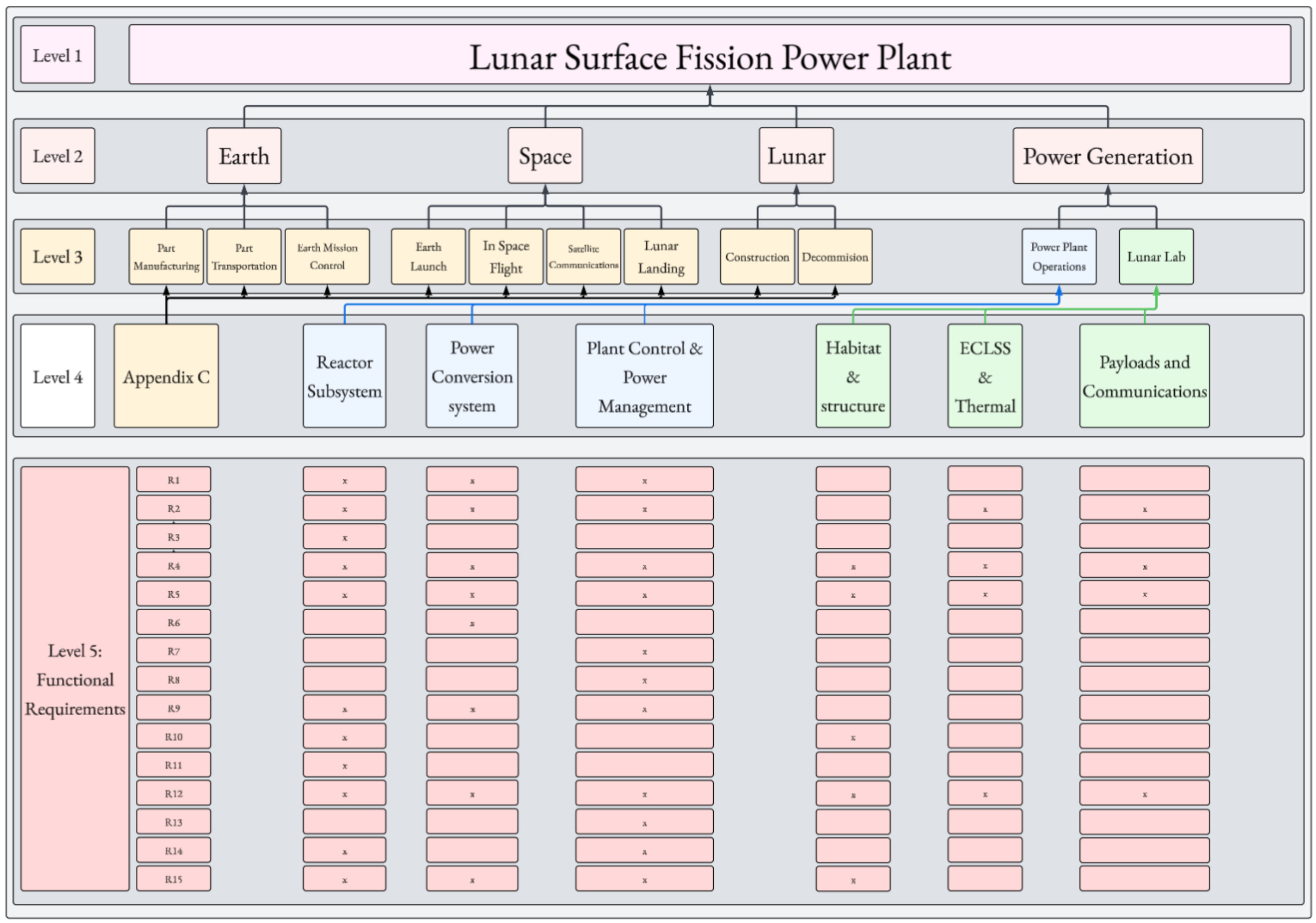
Functional Requirements

Req ID	Functional / Performance Requirement	Subsystem(s)
R1	System shall provide a continuous ≥ 40 kWe net electric output for 10 years of lunar operation.	Reactor Module / Power Conversior
R2	System shall operate autonomously for ≥ 12 months without crew intervention.	Reactor Control & Automation
R3	Reactor shall remain subcritical during launch and transit.	Launch Package / Reactor Core
R4	System shall survive launch vibrations up to 12 g and landing shock ≥ 25 g without loss of integrity.	Structural & Containment Subsysten
R5	Reactor, radiator, and power modules shall tolerate temperature swings from +120 °C to −130 °C.	Thermal Control / Radiators
R6	Radiator subsystem shall reject ≥ 200 kWh waste heat under vacuum and 1/6 g.	Radiator Panels / Loop Heat Pipes
R7	Power Management and Distribution (PMAD) shall transmit 40 kWe over 750 m with ≥ 95 % efficiency.	PMAD / Cables / Transformers
R8	System shall interface with energy storage units (batteries or solar) for load-balancing ≤ 5 % voltage ripple.	PMAD / Energy Storage
R9	Total reactor + radiator + PMAD mass shall not exceed 5,000 kg for a single lander deployment.	All Subsystems
R10	Reactor shielding shall limit dose ≤ 0.5 mSv/hr at 500 m distance to habitat.	Shielding / Regolith Barrier
R11	Reactor fuel shall use HALEU (19.75 % U-235) or equivalent enrichment for long-life criticality.	Reactor Core
R12	System shall achieve Technology Readiness Level ≥ 6 before launch qualification.	System Integration
R13	System shall maintain < 1 % frequency variation and < 5 % voltage ripple in delivered power.	Power Electronics
R14	End-of-life operations shall enable safe shutdown and containment for ≥ 100 years.	Reactor & Containment
R15	System design shall be deployable /constructed by robotic lander or rover without crew assembly.	Deployment Mechanism / Structure

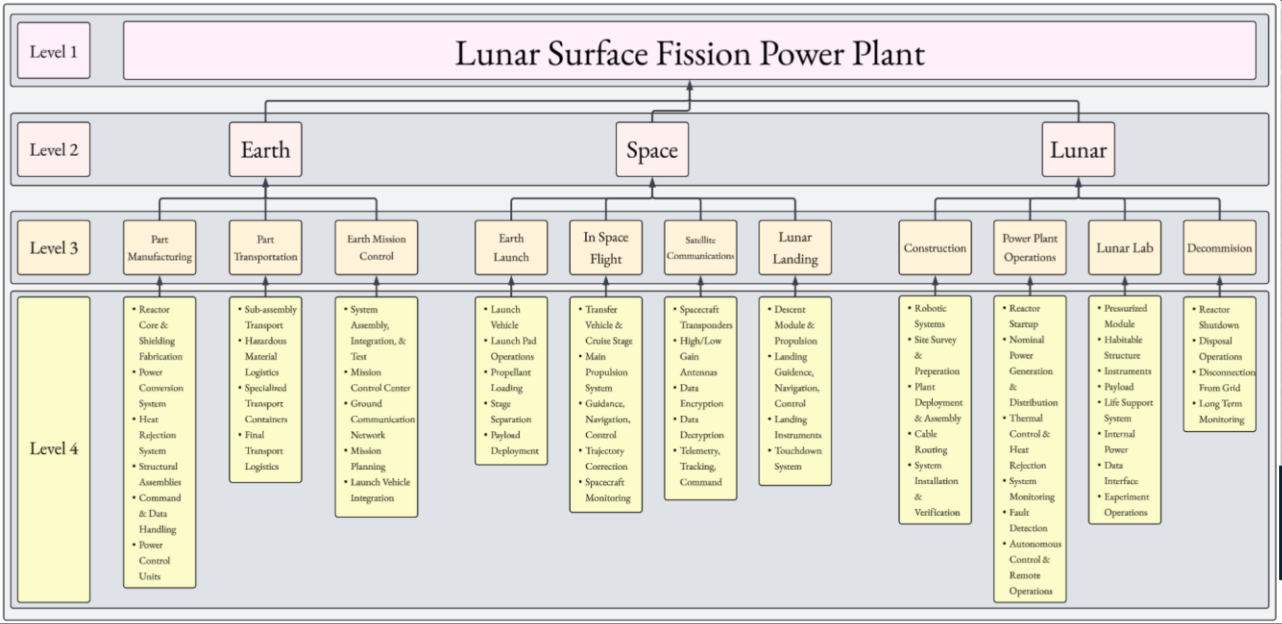
System Architecture (Segmented & Layered)



Design Drivers → System Architecture

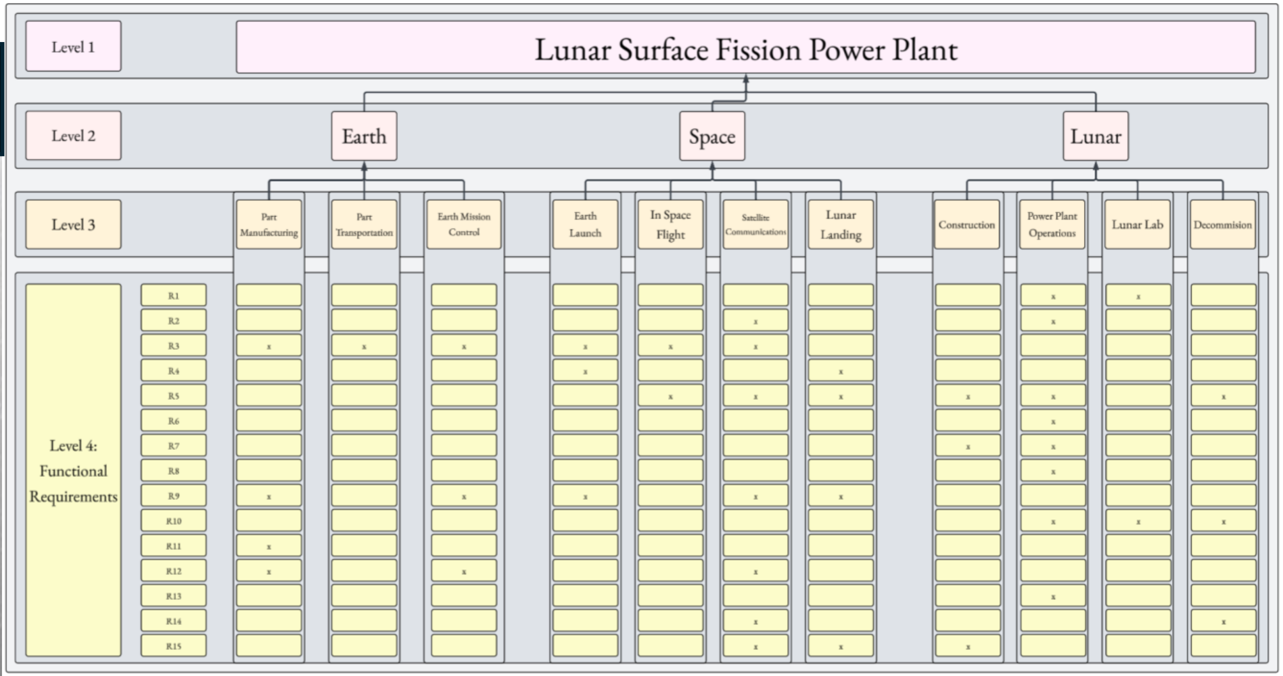


Additional Allocation Information



Appendix B

Appendix C



N^2 Diagram

Reactor Core		Thermal Energy (Heat)				Neutron Flux, Temperature Data		
Reactivity Change	Reactivity Control					Position Data		
Cooled Fluid In		Coolant & Pumps	Thermal Energy (Hot Fluid)			Pump Speed, Temperature, Pressure Data		
		Cooled Fluid Out	Power Conversion		Electircal Power	Engine Speed, Temperature, Power Data		
			Cooled Fluid Out	Waste Heat Rejection		Pump Speed, Temperature Data		
	Conditioned Power	Conditioned Power	Conditioned Power	Conditioned Power	PMAD	Conditioned Power	Conditioned Power	
					Power Status Data	Health & Status Sensor	Telemetry/Health Data	Telemetry/Health Data
	Position Commands	Pump Speed Commands	Engine Control Commands	Pump Speed Commands	Load Switching Commands		Autonomous Control	System Status and Alerts
							High Level Commands, Software Updates	Ground Control

Risk Assessment

Risk ID	Description	Probability (1–10)	Consequence (1–10)
Environmental/External			
R1	Lunar dust infiltration into systems	8	4
R2	Shielding degradation over time	6	6
R3	Spent fuel vault breach on Moon	2	10
R4	Rover or lander impact with reactor module or exposed cabling	2	9
R5	Interference between multiple nearby surface reactors or power systems	3	6
Power/Control			
R6	Reactor fails to start	3	8
R7	Control rod/mechanism failure	3	10
R8	Software control error at startup	2	6
R9	Power transmission failure to habitat	4	8
R10	Autonomous control algorithm fault during surface operation	3	9
R11	Radiation-induced electronics failure in control/PMAD units	5	8
R12	HVDC cable fracture from thermal cycling and mechanical stress	3	7
R13	Failure to fully insert or withdraw control rods on command	2	10
R14	Cybersecurity breach of remote command and telemetry link	3	8
R15	Extended loss of communications with plant and ground control	4	7
R16	Insufficient health monitoring sensors for autonomous fault detection	3	10

Risk ID	Description	Probability (1–10)	Consequence (1–10)
Regulatory/Programmatic			
R17	Launch approval delay or revocation due to regulatory concerns	7	5
R18	Failure to achieve TRL 6 prior to critical design/flight readiness	4	6
R19	Ground test anomaly requiring major redesign or delay	5	7
R20	Public or political opposition to nuclear launch impacting schedule	7	6
Structural/Mechanical			
R21	Lander crash on lunar surface	3	10
R22	Moonquake damage to containment	2	9
R23	Micrometeoroid strike on critical system	2	8
R24	Regolith berm construction failure leading to exceedance of dose limits	3	9
R25	Lander or deployment pallet tip-over during reactor offloading	4	10
R26	Fuel element cladding breach within core	2	10
Thermal			
R27	Failure to deploy radiator	4	8
R28	Loss of cooling loop performance	4	9
R29	Unexpected lunar thermal extremes	3	7
R30	Radiator freeze during low power operation	4	8
R31	Coolant boiling/voiding event due to local hot spots	2	8
R32	Progressive dust accumulation on radiators reducing heat rejection	6	7
R33	Radiator deployment misalignment reducing view factor to space	4	5
R34	Power conversion unit (Brayton/Stirling) degradation or failure	4	8
R35	Poor siting causing shadowing or obstruction of radiators	2	8

Most of our risks are rated with high consequences due to the difficulty of servicing the reactor on the Moon and the potential for a catastrophic nuclear disaster; however, these severe risks are relatively unlikely, as required by mission design. In contrast, certain risks, specifically some environmental and regulatory ones, tend to be more probable.

Risk Matrix

Justification

The green 'Acceptable' zone is small by design, forcing any risk with a non-trivial consequences (C > 4) into a 'Manageable' (Yellow) or 'Critical' (Red) category.

Likelihood \ Consequence	1: Little to None	2: Minimal	3: Minor	4: Noticeable	5: Moderate	6: Significant	7: Major	8: Severe	9: Critical	10: Catastrophic
10: Extremely Frequent										
9: Frequent										
8: Very Likely				R1						
7: Likely					R17	R20				
6: Occasional						R2	R32			
5: Possible							R19	R11		
4: Unlikely					R33	R18	R15	R9, R27, R30, R34	R28	R25
3: Rare						R5	R12, R29	R6, R14	R10, R24	R7, R16, R21
2: Very Rare						R8		R23, R31, R35	R4, R22	R3, R13, R26
1: Extremely Rare										

Technology Readiness & Development Needs

- Most of the technology is Earth verified but needs hardening and validation on the lunar surface
  - Lunar surface and atmospheric environment
  - Regolith build up and constant radiation flux
  - Need to add note about scaling reactor to 100 kwe
- However, there is **one** concerning area
  - Range and extent of AI autonomy in command
  - Can be easily replaced by PID and/or LQR controllers
- Implementation of PID/LQR controller to take over if AI fails

Technology Readiness & Development Needs

Area	TRL Estimate	Notes & Justification
Reactor Core (Fission)	6	Successfully demonstrated on Earth (KRUSTY/Pele) but has not operated in lunar conditions.
Brayton/Stirling Conversion	7	Advanced prototypes have run on Earth with long-duration data; lunar adaptation and vacuum operation remain to be proven.
Radiator Field Deployables	7	Deployable radiators demonstrated in relevant thermal vac tests but require validation under lunar dust and gravity conditions.
PMAD HVDC Trunk	5	Space-rated hardware exists, but lunar routing, temperature cycling, and regolith exposure lower maturity.
Autonomous FDIR Software	4	Algorithms under development; autonomy and fault recovery not yet flight-proven in lunar context.
Lunar Dust Mitigation (Seals/Covers)	4	Lab demonstrations exist, but no validation in lunar regolith environment or extended operation.
Radiation-Hardened Electronics	9	Proven for deep space and lunar missions; meets required dose tolerance for a 10-year mission.
10-Year Active Cooling Loop	5	Component-level maturity is high, but long-duration autonomous performance in lunar vacuum unverified.
Integrated Sensor Health Suite	6	Mature sensors exist; integration with autonomous diagnostics still maturing for lunar deployment.
Secure Remote Command Uplink	7	Flight-proven communication systems exist; requires integration and cybersecurity validation for nuclear payload.
Robotic Offloading System	6	Proven robotics and teleoperation technology, but lunar gravity and terrain introduce new failure modes.
Rocket Transportation	8	Orbital and lunar delivery vehicles are proven; minor adaptation needed for this payload and mission profile.
AI Integration	6	Demonstrated reliable performance and steady improvement in terrestrial applications; however, additional refinement and validation are needed, particularly due to limited AI training data and experience in lunar surface nuclear power environments.

Developmental Performance Metric

Autonomous FDIR/Artificial Intelligence Success Rate (%)

The Purpose

- Allows for constant tracking of R2
- Directly mitigates R8 & R10 (Autonomous algorithm faults)
- Tracks progress of our lowest TRL item, FDIR

The Metric

- FDIR Success Rate (%): The percentage of simulated abnormal scenarios the system can successfully Detect, Isolate, and Recover from
- Measured with a high-confidence simulation using a "digital twin" of the system

The Developmental Goal

- Phase 1
  - 90% success on 50 simple, known faults (e.g., "Pump A failed").
- Phase 2
  - 95% success on 200+ complex, cascading faults (e.g., "Sensor 1 failed, causing Pump A to cycle, causing thermal spike").
- Phase 3
  - 99.9% success rate
  - Mean time to recovery < 15 seconds
  - Zero false positives

Operational Performance Metrics

Primary Metric: System Availability (%)

The Purpose

- Measures long-term reliability of plant operations
- Ensures continuous delivery of ≥ 40 kWe to mission loads
- Enables trend tracking and design for availability decisions

The Metric

- System Availability (%) = time delivering ≥ 40 kWe ÷ total mission time
- Downtime includes maintenance, faults, startup transients, or control interruptions
- A 40 kWe plant that is "up" only 50% of the time is considered a failure

The Developmental Goal

- Phase 1
  - 97.0% availability averaged over 10 years
- Phase 2
  - 98.0% availability averaged over 10 years
- Phase 3
  - 99.0% availability averaged over 10 years

Operational Performance Metrics

Autonomy Metric: Human Intervention Rate

The Purpose

- Distinguishes “it’s on” from “it’s on by itself” (supports R2/R10)
- Quantifies operations burden on Ground Control
- Drives fault-tolerant design and autonomous recovery

The Metric

- Unplanned remote interventions per year from Ground Control
- Count an event when human action changes system state to recover or protect power
- Excludes planned tests/training and scheduled maintenance

The Operational Goal

- R2: ≤ 12 per year (≤ 1/month)
- R10: ≤ 2 per year, trending to 0 in steady state
- Report mean time to recovery and root-cause category for each intervention

Operational Performance Metrics

Quality Metric: Power Stability

The Purpose

- Tracks PMAD and power-conversion health
- Protects mission equipment from “dirty” power
- Ensures stable delivery to sensitive loads

The Metric

- Voltage ripple: < 2% (RMS, specified bandwidth)
- Frequency variation: < 1% (steady state; approved transients excluded)
- Measured at the delivery bus under representative load

The Operational Goal

- Target: Maintain 100% compliance with R13 during nominal and contingency operations
- Log and disposition any exceptions (duration, magnitude, cause)

Lesson's Learned

Early definitions of interfaces is critical

Creating realistic performance metrics improves project maturity

Harsh lunar conditions changes priorities

Scalability relies on modular control

Prioritizing depth improves clarity

High consequence risk requires redundant mitigation

Earth-based validation isn't enough

THANK YOU, QUESTIONS?

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